

Actividad 3 Algoritmia y programación

Hecho por:

Keller Acevedo

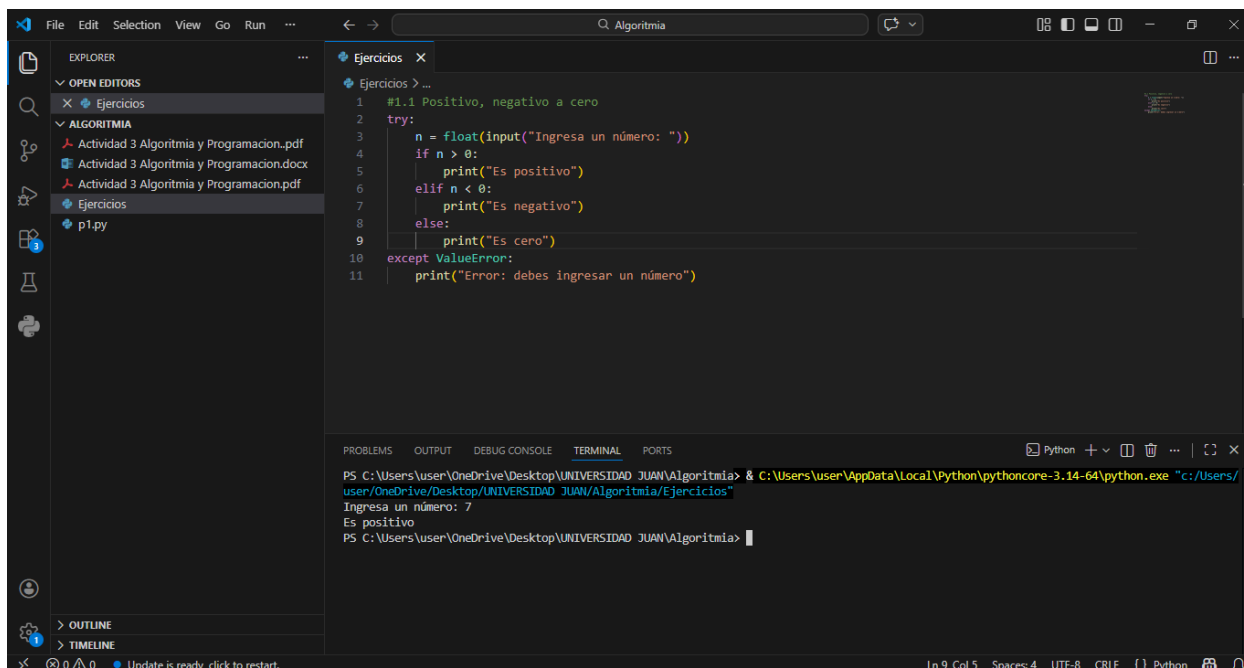
Universidad Minuto de Dios

Algoritmia y programación

Docente: Ingeniera Carolina Ulloa

BOGOTÁ

Abril 2026

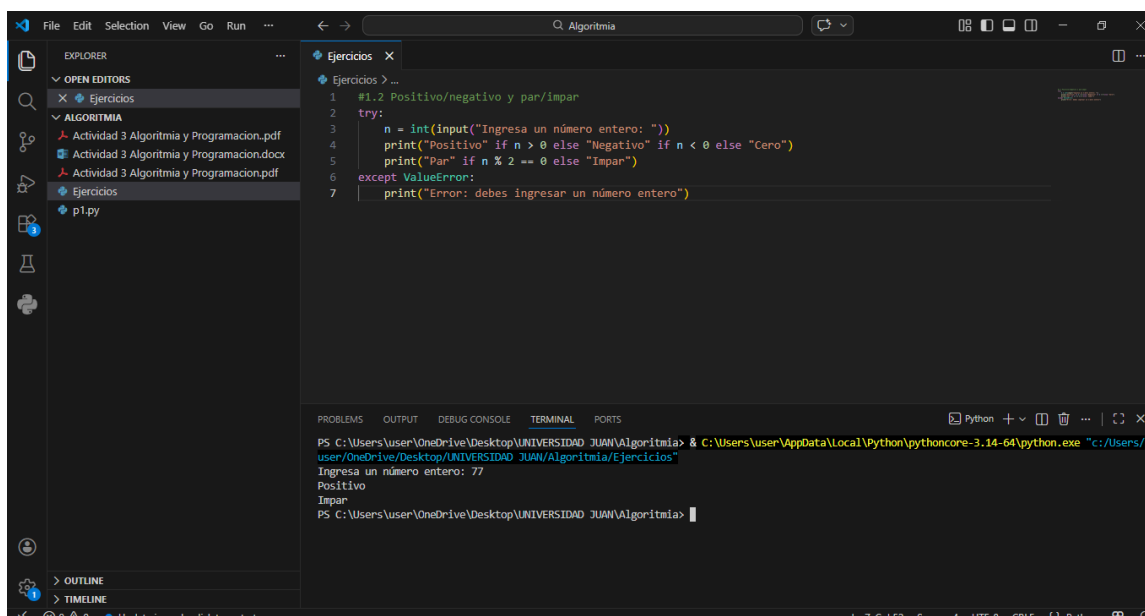


The screenshot shows the Visual Studio Code interface with the Explorer sidebar on the left. The 'Ejercicios' folder is selected, showing a file named 'p1.py'. The main editor window displays the code for 'Ejercicios > ...'.

```
1 #1.1 Positivo, negativo a cero
2 try:
3     n = float(input("Ingresa un número: "))
4     if n > 0:
5         print("Es positivo")
6     elif n < 0:
7         print("Es negativo")
8     else:
9         print("Es cero")
10 except ValueError:
11     print("Error: debes ingresar un número")
```

The terminal at the bottom shows the command to run the script and the output:

```
PS C:\Users\user\OneDrive\Desktop\UNIVERSIDAD JUAN\Algoritmia> & C:\Users\user\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/user/OneDrive/Desktop/UNIVERSIDAD JUAN/Algoritmia/Ejercicios"
Ingresa un número: 7
Es positivo
PS C:\Users\user\OneDrive\Desktop\UNIVERSIDAD JUAN\Algoritmia>
```



The screenshot shows the Visual Studio Code interface with the Explorer sidebar on the left. The 'Ejercicios' folder is selected, showing a file named 'p1.py'. The main editor window displays the code for 'Ejercicios > ...'.

```
1 #1.2 Positivo/negativo y par/impar
2 try:
3     n = int(input("Ingresa un número entero: "))
4     print("Positivo" if n > 0 else "Negativo" if n < 0 else "Cero")
5     print("Par" if n % 2 == 0 else "Impar")
6 except ValueError:
7     print("Error: debes ingresar un número entero")
```

The terminal at the bottom shows the command to run the script and the output:

```
PS C:\Users\user\OneDrive\Desktop\UNIVERSIDAD JUAN\Algoritmia> & C:\Users\user\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/user/OneDrive/Desktop/UNIVERSIDAD JUAN/Algoritmia/Ejercicios"
Ingresa un número entero: 77
Positivo
Impar
PS C:\Users\user\OneDrive\Desktop\UNIVERSIDAD JUAN\Algoritmia>
```

The screenshot shows the Visual Studio Code interface with a file explorer on the left and a code editor on the right. The file explorer shows a project named 'Ejercicios' with a file 'p1.py'. The code editor displays a Python script titled '#1.3 Mayor de tres números'. The script uses a try-except block to handle user input for three numbers and prints the maximum. The terminal at the bottom shows the command to run the script and the resulting output.

```
1 #1.3 Mayor de tres números
2
3
4 try:
5     a = float(input("Primer número: "))
6     b = float(input("Segundo número: "))
7     c = float(input("Tercer número: "))
8     mayor = a
9     if b > mayor:
10         mayor = b
11     if c > mayor:
12         mayor = c
13     print(f"El mayor es: {mayor}")
14 except ValueError:
15     print("Error: ingresa solo números")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\user\OneDrive\Desktop\UNIVERSIDAD JUAN\Algoritmia> & C:\Users\user\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/user/OneDrive/Desktop/UNIVERSIDAD JUAN/Algoritmia/Ejercicios"
Primer número: 2
Segundo número: 5
Tercer número: 1
El mayor es: 5.0
PS C:\Users\user\OneDrive\Desktop\UNIVERSIDAD JUAN\Algoritmia>
```

The screenshot shows the Visual Studio Code interface with a file explorer on the left and a code editor on the right. The file explorer shows a project named 'Ejercicios' with a file 'p1.py'. The code editor displays a Python script titled '#1.4 Grupo por edad'. The script uses a try-except block to handle user input for age and prints the corresponding group. The terminal at the bottom shows the command to run the script and the resulting output.

```
1 #1.4 Grupo por edad
2
3
4 try:
5     edad = int(input("Ingresa tu edad: "))
6     if edad < 0:
7         print("Edad no válida")
8     elif edad <= 12:
9         print("Niño")
10    elif edad <= 17:
11        print("Adolescente")
12    elif edad <= 59:
13        print("Adulto")
14    else:
15        print("Adulto mayor")
16 except ValueError:
17    print("Error: ingresa un número entero")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\user\OneDrive\Desktop\UNIVERSIDAD JUAN\Algoritmia> & C:\Users\user\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/user/OneDrive/Desktop/UNIVERSIDAD JUAN/Algoritmia/Ejercicios"
Ingresa tu edad: 18
Adulto
PS C:\Users\user\OneDrive\Desktop\UNIVERSIDAD JUAN\Algoritmia>
```

The screenshot shows the Visual Studio Code interface with a file explorer on the left and a code editor on the right. The file explorer shows a project named 'Ejercicios' with a file 'p1.py'. The code editor displays a Python script for comparing two numbers. The terminal at the bottom shows the command to run the script and its output.

```
1 # 1.5 Comparar dos números
2
3
4 try:
5     a = float(input("Primer número: "))
6     b = float(input("Segundo número: "))
7     if a == b:
8         print("Son iguales")
9     elif a > b:
10        print(f"{a} es mayor")
11    else:
12        print(f"{b} es mayor")
13 except ValueError:
14    print("Error: ingresa solo números")
```

Terminal output:

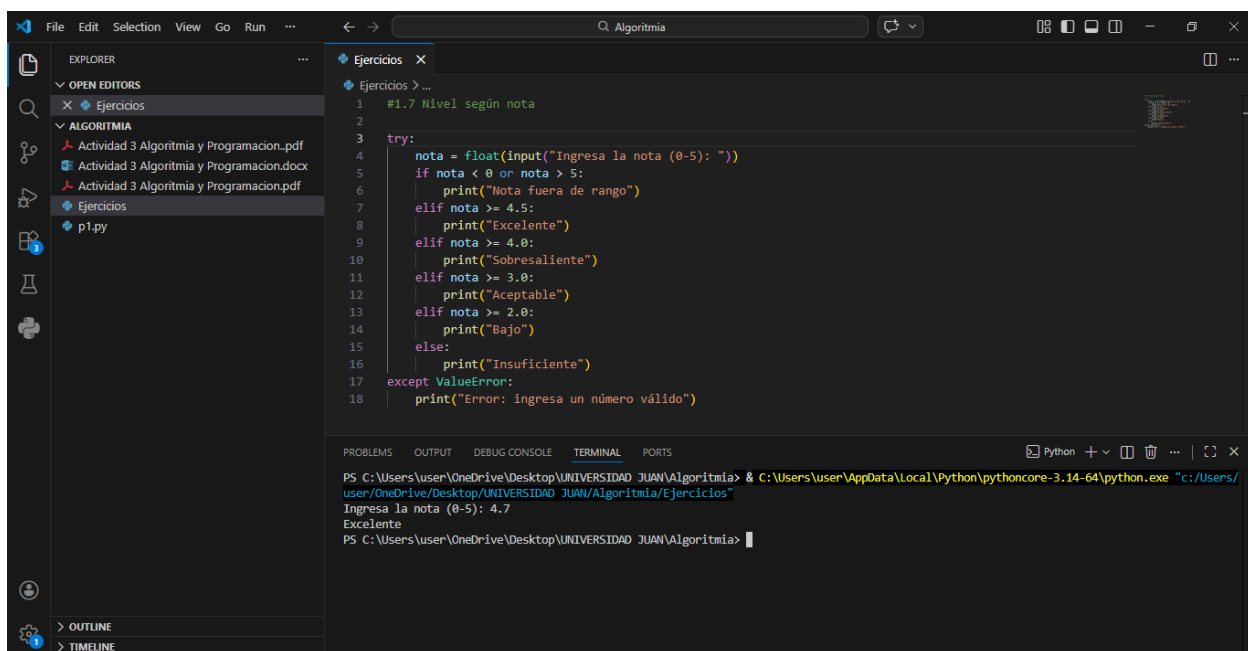
```
PS C:\Users\user\OneDrive\Desktop\UNIVERSIDAD JUAN\Algoritmia> & C:\Users\user\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/user/OneDrive/Desktop/UNIVERSIDAD JUAN/Algoritmia/Ejercicios/p1.py"
Primer número: 7
Segundo número: 77
77.0 es mayor
PS C:\Users\user\OneDrive\Desktop\UNIVERSIDAD JUAN\Algoritmia>
```

The screenshot shows the Visual Studio Code interface with a file explorer on the left and a code editor on the right. The file explorer shows a project named 'Ejercicios' with a file 'p1.py'. The code editor displays a Python script for validating a password. The terminal at the bottom shows the command to run the script and its output.

```
1 #1.6 Validar contraseña
2
3 CONTRASENA = "Tarea3"
4 clave = input("Ingresa la contraseña: ")
5 if clave == CONTRASENA:
6     print("Acceso permitido")
7 else:
8     print("Acceso denegado")
```

Terminal output:

```
PS C:\Users\user\OneDrive\Desktop\UNIVERSIDAD JUAN\Algoritmia> & C:\Users\user\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/user/OneDrive/Desktop/UNIVERSIDAD JUAN/Algoritmia/Ejercicios/p1.py"
Ingresa la contraseña: Tarea5
Acceso denegado
PS C:\Users\user\OneDrive\Desktop\UNIVERSIDAD JUAN\Algoritmia>
```

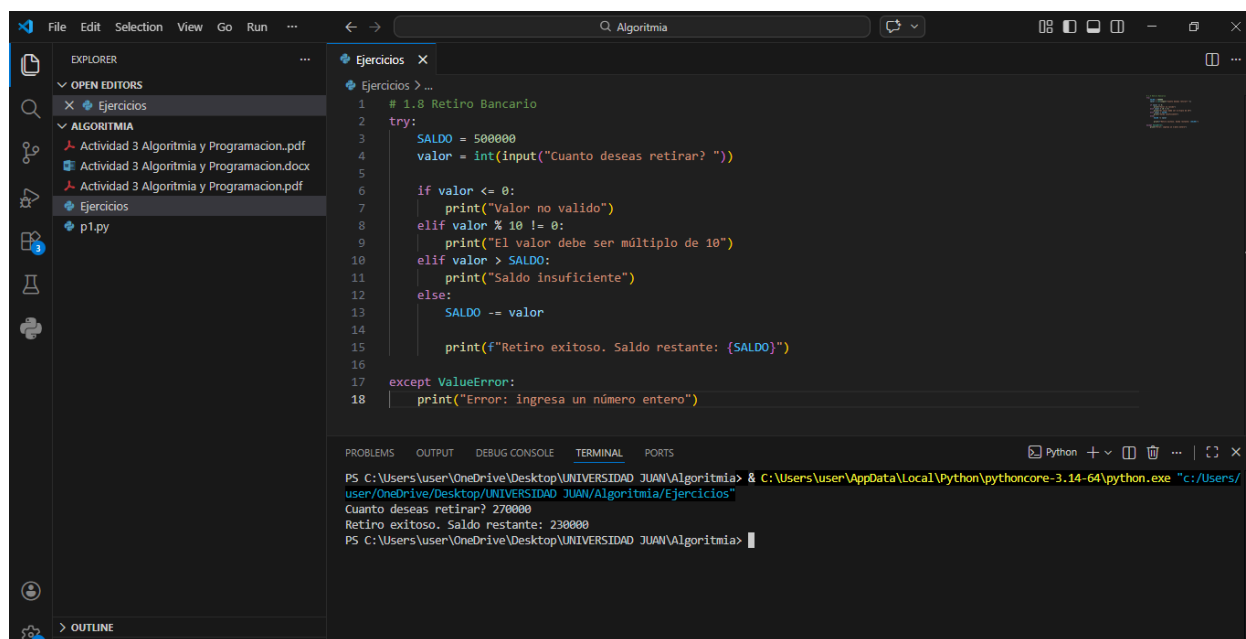


The screenshot shows the Visual Studio Code interface with a file explorer on the left containing a folder named 'Ejercicios' with a file 'p1.py'. The main editor displays a Python script for calculating a grade level based on a score. The terminal at the bottom shows the command to run the script and the resulting output.

```
1 #1.7 Nivel según nota
2
3 try:
4     nota = float(input("Ingresa la nota (0-5): "))
5     if nota < 0 or nota > 5:
6         print("Nota fuera de rango")
7     elif nota >= 4.5:
8         print("Excelente")
9     elif nota >= 4.0:
10        print("Sobresaliente")
11    elif nota >= 3.0:
12        print("Aceptable")
13    elif nota >= 2.0:
14        print("Bajo")
15    else:
16        print("Insuficiente")
17 except ValueError:
18    print("Error: ingresa un número válido")
```

Terminal output:

```
PS C:\Users\user\OneDrive\Desktop\UNIVERSIDAD JUAN\Algoritmia> & C:\Users\user\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/user/OneDrive/Desktop/UNIVERSIDAD JUAN/Algoritmia/Ejercicios"
Ingresa la nota (0-5): 4.7
Excelente
PS C:\Users\user\OneDrive\Desktop\UNIVERSIDAD JUAN\Algoritmia>
```

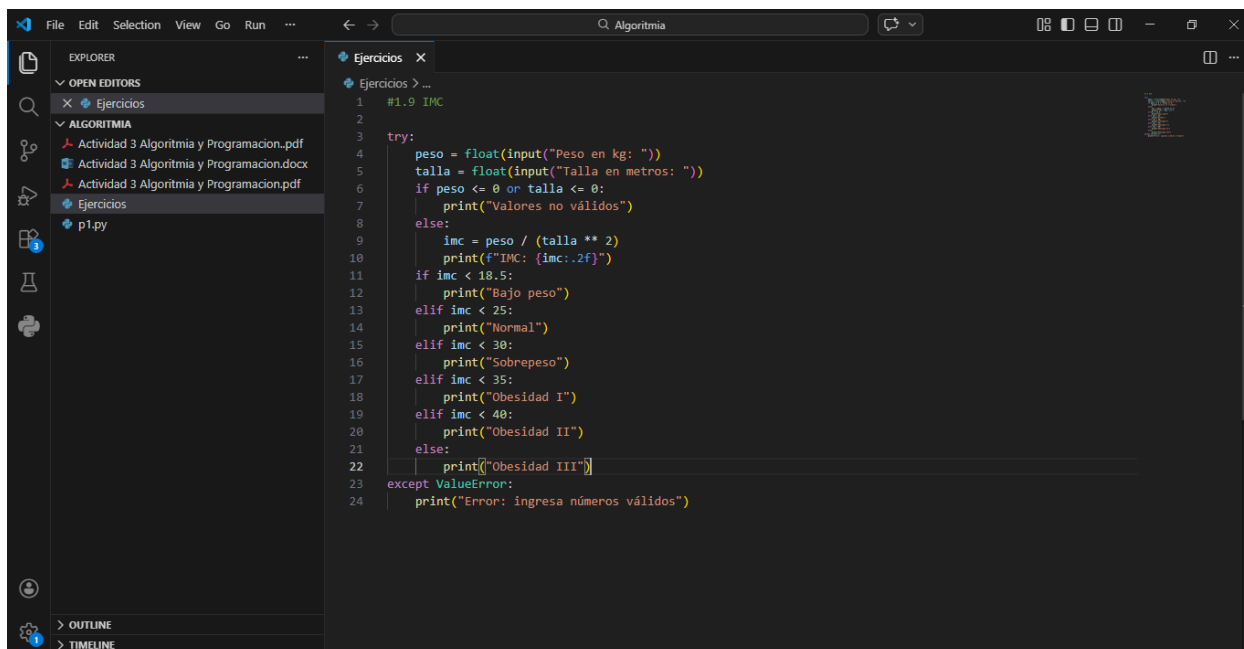


The screenshot shows the Visual Studio Code interface with a file explorer on the left containing a folder named 'Ejercicios' with a file 'p1.py'. The main editor displays a Python script for a bank withdrawal simulation. The terminal at the bottom shows the command to run the script and the resulting output.

```
1 # 1.8 Retiro Bancario
2 try:
3     SALDO = 500000
4     valor = int(input("Cuanto deseas retirar? "))
5
6     if valor <= 0:
7         print("Valor no valido")
8     elif valor % 10 != 0:
9         print("El valor debe ser múltiplo de 10")
10    elif valor > SALDO:
11        print("Saldo insuficiente")
12    else:
13        SALDO -= valor
14
15    print(f"Retiro exitoso. Saldo restante: {SALDO}")
16
17 except ValueError:
18    print("Error: ingresa un número entero")
```

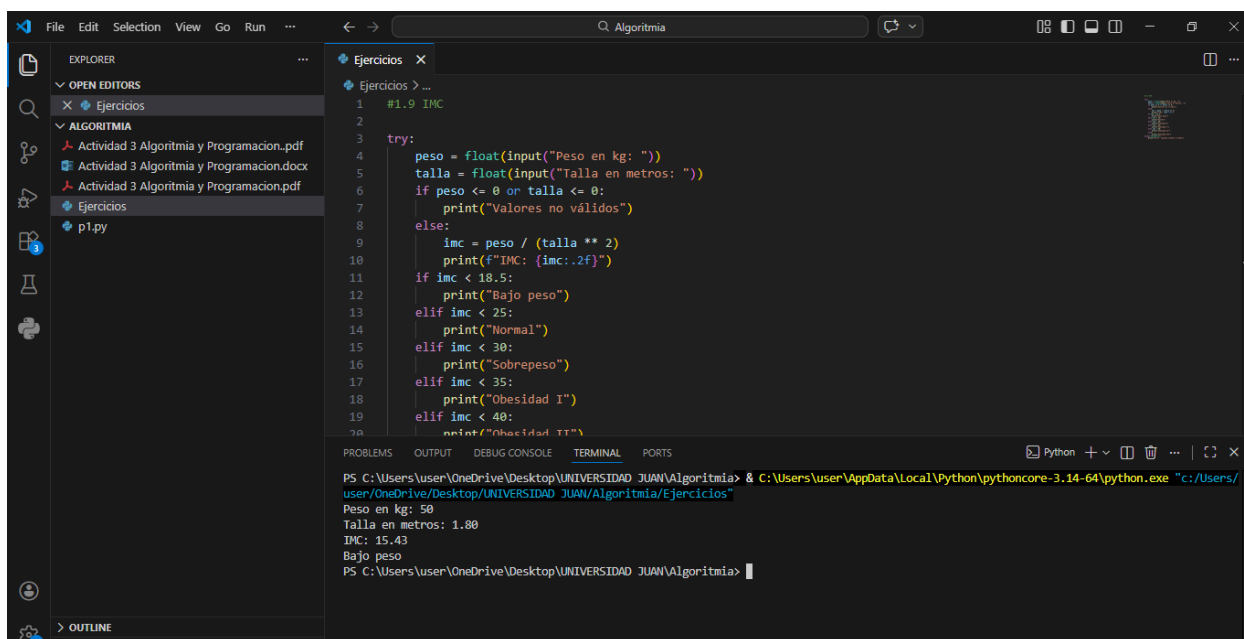
Terminal output:

```
PS C:\Users\user\OneDrive\Desktop\UNIVERSIDAD JUAN\Algoritmia> & C:\Users\user\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/user/OneDrive/Desktop/UNIVERSIDAD JUAN/Algoritmia/Ejercicios"
Cuanto deseas retirar? 270000
Retiro exitoso, Saldo restante: 230000
PS C:\Users\user\OneDrive\Desktop\UNIVERSIDAD JUAN\Algoritmia>
```



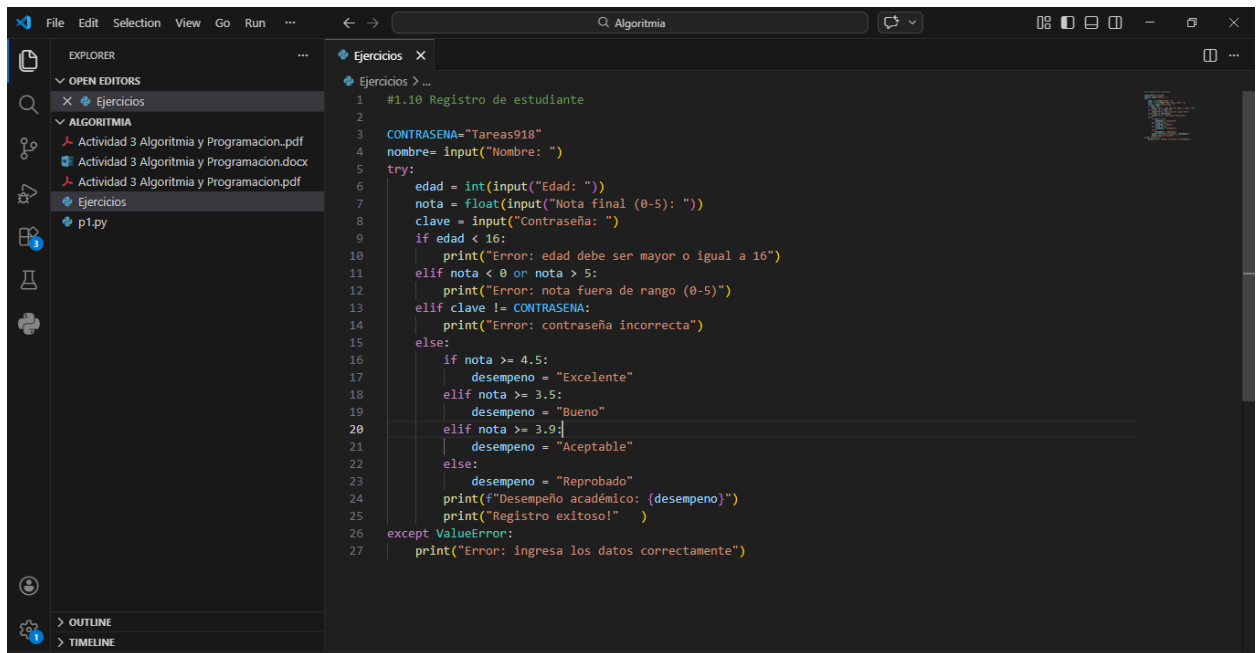
The screenshot shows the Visual Studio Code editor with a file named `p1.py` open. The code is a Python script for calculating BMI. It uses `try-except` blocks to handle `ValueError` exceptions. The script prompts the user for weight in kg and height in meters, then calculates the BMI and prints a message based on the result.

```
1 #1.9 IMC
2
3 try:
4     peso = float(input("Peso en kg: "))
5     talla = float(input("Talla en metros: "))
6     if peso <= 0 or talla <= 0:
7         print("Valores no válidos")
8     else:
9         imc = peso / (talla ** 2)
10        print(f"IMC: {imc:.2f}")
11    if imc < 18.5:
12        print("Bajo peso")
13    elif imc < 25:
14        print("Normal")
15    elif imc < 30:
16        print("Sobrepeso")
17    elif imc < 35:
18        print("Obesidad I")
19    elif imc < 40:
20        print("Obesidad II")
21    else:
22        print("Obesidad III")
23 except ValueError:
24    print("Error: ingresa números válidos")
```



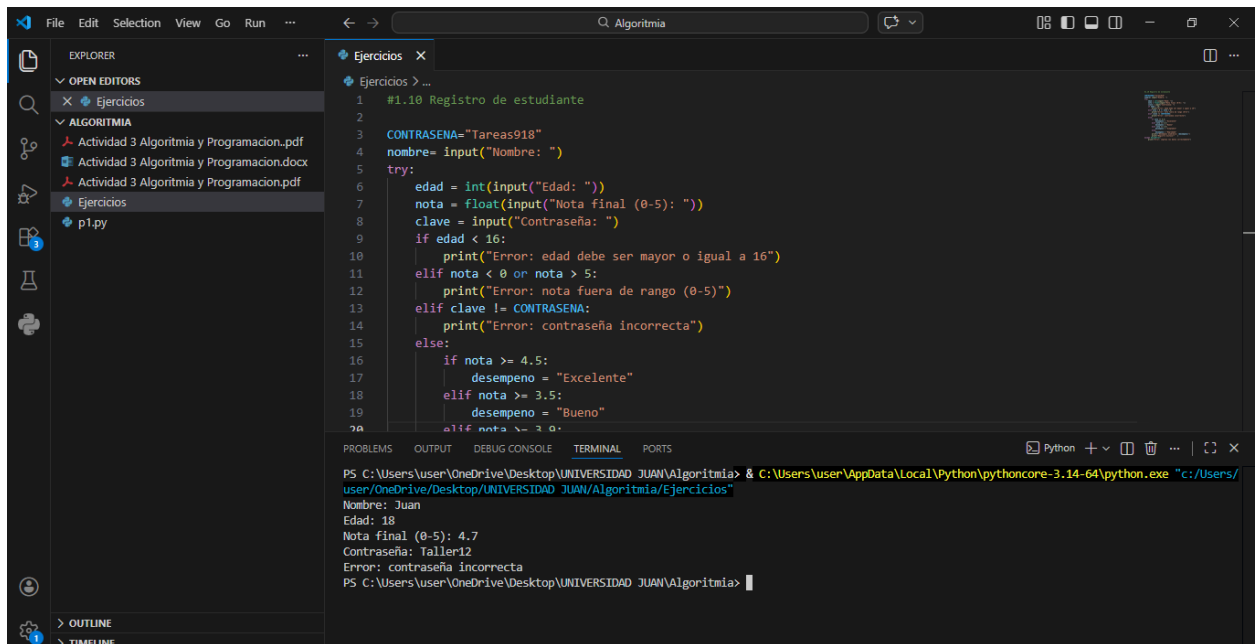
The screenshot shows the same VS Code editor with the `p1.py` file. The `TERMINAL` panel at the bottom shows the command prompt output of running the script. The user has entered a weight of 50 kg and a height of 1.80 meters, resulting in a BMI of 15.43, which is categorized as "Bajo peso".

```
PS C:\Users\user\OneDrive\Desktop\UNIVERSIDAD JUAN\Algoritmia> & C:\Users\user\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/user/OneDrive/Desktop/UNIVERSIDAD JUAN/Algoritmia/Ejercicios"
Peso en kg: 50
Talla en metros: 1.80
IMC: 15.43
Bajo peso
PS C:\Users\user\OneDrive\Desktop\UNIVERSIDAD JUAN\Algoritmia>
```



The screenshot shows the Visual Studio Code editor with a file named `p1.py` open. The code is a Python script for student registration. It includes a try-except block to handle user input errors. The script prompts for a name, age, final grade, and password. It then checks if the age is at least 16, if the grade is between 0 and 5, and if the password matches the predefined `CONTRASENA`. Based on the grade, it assigns a performance level: "Excelente" (4.5 and above), "Bueno" (3.5 and above), "Aceptable" (3.0 and above), or "Reprobado" (below 3.0). It prints the grade and performance level, and a success message. If there's a `ValueError`, it prints an error message.

```
1 #1.10 Registro de estudiante
2
3 CONTRASENA="Tareas918"
4 nombre= input("Nombre: ")
5 try:
6     edad = int(input("Edad: "))
7     nota = float(input("Nota final (0-5): "))
8     clave = input("Contraseña: ")
9     if edad < 16:
10         print("Error: edad debe ser mayor o igual a 16")
11     elif nota < 0 or nota > 5:
12         print("Error: nota fuera de rango (0-5)")
13     elif clave != CONTRASENA:
14         print("Error: contraseña incorrecta")
15     else:
16         if nota >= 4.5:
17             desempeno = "Excelente"
18         elif nota >= 3.5:
19             desempeno = "Bueno"
20         elif nota >= 3.0:
21             desempeno = "Aceptable"
22         else:
23             desempeno = "Reprobado"
24         print(f"Desempeño académico: {desempeno}")
25         print("Registro exitoso!")
26 except ValueError:
27     print("Error: ingresa los datos correctamente")
```



The screenshot shows the same VS Code editor with the terminal window open at the bottom. The terminal displays the output of the Python script. The user has entered the name "Juan", age "18", grade "4.7", and password "Taller12". The script outputs the grade and performance level, and a success message. The terminal prompt is `PS C:\Users\user\OneDrive\Desktop\UNIVERSIDAD JUAN\Algoritmia>`.

```
PS C:\Users\user\OneDrive\Desktop\UNIVERSIDAD JUAN\Algoritmia> & C:\Users\user\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/user/OneDrive/Desktop/UNIVERSIDAD JUAN/Algoritmia/Ejercicios"
Nombre: Juan
Edad: 18
Nota final (0-5): 4.7
Contraseña: Taller12
Error: contraseña incorrecta
PS C:\Users\user\OneDrive\Desktop\UNIVERSIDAD JUAN\Algoritmia>
```